FinRL stock selection: data repair and fixed-candidate evaluation

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The predeclared top-10% candidate exceeds SPY on the available full trading window, 2020-06-01 to 2025-09-30: 175.22% versus 135.84% cumulative return after strategy trading costs. This is a coverage-limited historical result. Missing failed/acquired-company histories and unverified original statement vintages still prevent a claim that the sample is free of survivorship or look-ahead bias.

1. What changed and why

The course top-25% rule remains the baseline. The proposed submission rule is top 10%, chosen using the 7 October development results and fixed before this repaired run. We do not treat the new run or 2024 onward as an untouched holdout, and do not change the threshold after seeing this run.

Change	Evidence / implication
Price coverage	47 previously missing ticker series accepted using current SEC/Yahoo identity checks, complete in-life adjusted prices and, where available, overlapping return checks. 59 original missing symbols remain unresolved or excluded.
Ticker identity	Official same-issuer changes and share-class notation use a canonical security identifier. One financial observation and one complete price source per security avoid duplicate counting and mixed-provider series.
Financial joins	Raw payloads now match the aggregate income filing date, then the actual fiscal period. Treating fiscal Q1 as calendar Q1 matched the wrong records for several firms.
Financial spot checks	48 fixed sample rows; 30 had raw payloads. EPS matched 30/30 and current ratio 29/29 after correcting the join. Quarter-price / quarterly-EPS explains the supplied PE definition.
Trade accounting	Quarterly close trades, next-session return accrual, drifted holdings and 10 bps per dollar bought/sold. A missing held quote raises an error; no indefinite forward-fill or invented terminal proceeds.

2. Data and model protocol

The prepared inputs contain 22,909 rows. Within the price window the historical membership filter changes 21,601 rows / 701 tickers to 18,772 rows / 684 tickers. Timely quoted inputs contain 16,454 rows before rolling model eligibility; final predictions contain 10,204 rows.

RF, XGBoost and LightGBM use the same four fixed parameter candidates per model and seed 42. Expanding chronological training folds fit median imputation only on the training fold. Completed-outcome purges separate training, outer validation and inference. Lowest outer-validation MSE chooses the model for each sector/quarter. The 52 financial features exclude absolute adjusted-price levels. No extra hyperparameter search was introduced.

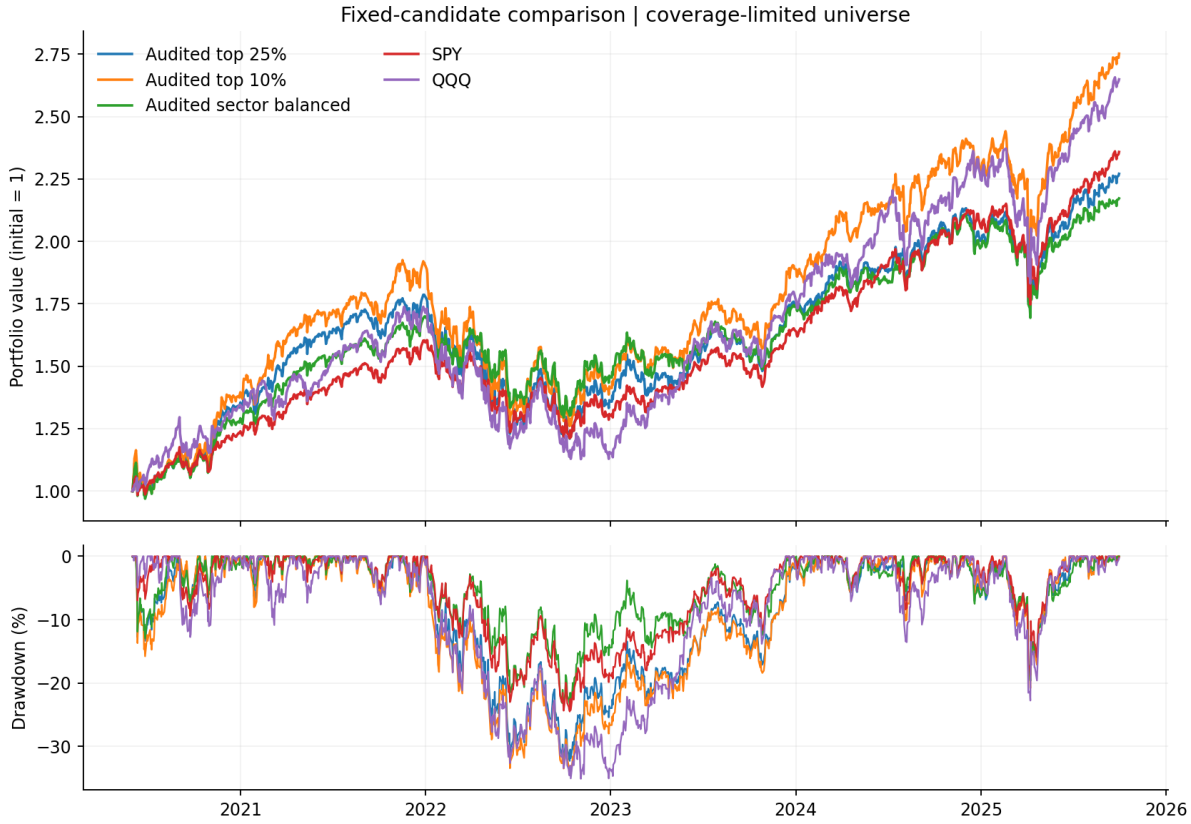
3. Full trading window and the original common window

Full available window: 2020-06-01 to 2025-09-30, 1,341 trading dates. The first day is the earliest saved audited inference, rather than the latest start among older comparison versions.

Portfolio	Cumulative	CAGR	Vol.	Sharpe*	Max DD	TO/year
Top 10% (fixed candidate)	175.22%	20.92%	22.92%	0.94	-34.54%	4.85x
Top 25% rule baseline	127.08%	16.63%	20.27%	0.86	-32.29%	3.81x
Sector-balanced top 25%	117.23%	15.67%	18.22%	0.89	-23.82%	3.97x
All model candidates, equal	112.73%	15.21%	17.18%	0.91	-20.37%	0.78x
Quoted index members, equal	112.59%	15.20%	17.20%	0.91	-20.74%	0.62x
SPY	135.84%	17.46%	17.47%	1.01	-24.50%	0.00x
QQQ	164.92%	20.05%	23.06%	0.91	-35.12%	0.00x

* Daily Sharpe uses zero risk-free rate and 252-day annualization. Turnover counts both buys and sells. SPY/QQQ use adjusted closes over identical dates, no explicit benchmark trading fee, with fund expenses embedded in observed prices. These are equity backtests without leverage.

Original common interval	Top 10%	Top 25%	SPY	QQQ
2020-09-01 to 2025-09-30	139.72%	100.02%	103.47%	106.38%



Both tables use the repaired data. The 7 October results are retained separately and must not be presented as an isolated causal comparison: data coverage, fiscal matching, security identity and price-source choices changed together.

4. What drove the candidate result

Contributions below are dollar PnL divided by initial portfolio NAV, summed through time. They are additive percentage points of the final NAV change, not each stock's buy-and-hold return or a causal estimate of alpha.

Largest stock contributions	Sector	Contribution
NVDA	Technology	9.54% pts
ANET	Technology	8.53% pts
AVGO	Technology	6.87% pts
CFG	Financial Services	6.18% pts
PLTR	Technology	5.70% pts
GLW	Technology	5.55% pts

Sector	Contribution
Technology	73.18% pts
Financial Services	54.90% pts
Industrials	31.60% pts
Energy	7.54% pts
Real Estate	4.85% pts
Consumer Cyclical	3.60% pts

All stock contributions sum to 179.70% of initial NAV. Trading costs remove 4.49%; the reconciled net change is 175.22%. Sector names are retrospective classifications and require that qualification.

Model comparison and feature importance

Model	Wins	Mean val. MSE	Top normalized importances
Random Forest	173	0.021871	BPS, revenue_per_share, capex_per_share
XGBoost	39	0.023334	revenue_per_share, debt_to_mktcap, ps
LightGBM	30	0.023889	peg, effective_tax_rate, debt_service_coverage

Importances are normalized within each fitted model/window. RF impurity, XGBoost gain and LightGBM split counts differ; their magnitudes are not directly comparable. Validation MSE wins are not evidence of statistically significant superiority.

Uncertainty and disclosure delay

Top-10% annualized arithmetic excess return over SPY is 4.00%. The descriptive 63-day block-bootstrap 95% interval is -2.67% to 10.58%. This conditions on the dataset and excludes strategy-selection and missing-data uncertainty; statistical significance is not an added course requirement.

Requiring five extra business days of disclosure lead time, while keeping forecasts fixed, retains 9,463/10,204 predictions and yields 176.23% cumulative return. This is an eligibility sensitivity, not a separately retrained model.

5. Costs, failures and remaining work

Portfolio / delay	5 bps	10 bps	20 bps	50 bps
Top 10% / 0 sessions	21.21%	20.92%	20.33%	18.59%
Top 10% / 1 sessions	20.29%	20.00%	19.42%	17.70%
Top 10% / 5 sessions	18.40%	18.12%	17.55%	15.85%
Top 25% / 0 sessions	16.85%	16.63%	16.19%	14.87%
Top 25% / 1 sessions	16.46%	16.24%	15.80%	14.48%
Top 25% / 5 sessions	14.21%	13.99%	13.56%	12.27%

Entries are annualized returns over the full window. Delayed variants begin from cash and use then-current membership and executable quotes; they do not credit the entry-day move.

What did not work

The original notebook reported 27.16% cumulative return while combining inconsistent stock/benchmark return conventions and missing active-stock returns. Its headline return is retained as reproduction history rather than a comparable investment score. The first raw-date audit used fiscal/calendar quarter keys that did not consistently identify the aggregate statement. Partial public quote downloads could not safely restore failed-bank, share-class-change or merger outcomes; uncertain series were excluded instead of assigning terminal value zero or treating stale quotes as realizable cash.

Data gaps that remain

At the first audited coverage date, 450/505 historical members have quotes; at the last coverage date, 502/503. The decision ledger records every accepted and rejected recovery. Missing examples include SIVB, FRC, SBNY, TWTR and ATVI. We have no certified stockholder terminal proceeds for these names. Therefore the graded survivorship-free sample requirement remains unresolved even where the observed return exceeds SPY.

Current SEC identity checks and provider-adjusted histories do not prove every historical security mapping or corporate-action adjustment. Financial filing-date checks establish recorded availability within the supplied snapshot, not original unrevised financial vintages. SEC/source coverage is missing for part of the sample. No real fill, market-impact or capacity study was performed.

Reproduction and submission status

README gives one end-to-end command for source verification, recovery, financial checks, training, both windows, numerical checks and PDF creation. The verifier takes arbitrary run/comparison paths; sample_flow is produced during training. Mutable public recovery prices can change, so their retrieval metadata, source hashes and return-consistency checks must accompany reproduced results.

Patrick Xiong (sx2434). Medium article published on 9 October; link in README. The course PR is a draft while historical data repairs remain pending WRDS approval. Numerical results on these pages are the preserved 8 October experiment; they have not been rerun on WRDS data. Code, report and derived outputs are packaged separately from raw inputs.

Sources

Course instructions; pinned course dataset; SEC company tickers; Yahoo Finance adjusted-price chart histories; official corporate-action references in evidence/root_corporate_action_sources.json. Failed-bank context: FDIC 2023 failures. These sources do not supply a complete historical equity-delisting-return dataset.